



WHITING SCHOOL OF ENGINEERING

PATIENT SAFETY COLLABORATION

CLASS: STRATEGIES IN INNOVATION & GROWTH

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COMPANY: THE JOHNS HOPKINS MEDICINE



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Executive Summary

The patient flow process at Bayview Medical Center, from triage to discharge and bed management, faces significant challenges in managing surges effectively. Historically, Active Bed Management (ABM) has shown positive outcomes in controlling surges, but recent trends highlight an increase in ambulatory bypass instances, leading to bottlenecks in patient throughput. Despite improvements like electronic health records, the current predictive model relies heavily on trailing indicators, such as bed occupancy, leading to delayed responses to surges.

The Medicine Department, accounting for 50% of the hospital's bed capacity, plays a critical role in addressing these issues. The department faces unpredictable surges driven by inflows from the Emergency Department (ED), interdepartmental transfers, and external referrals. As a result, incoming patients from the ED often face extended wait times due to capacity constraints, which negatively impacts patient care and increases the risk of ambulance bypass. Recognizing these inefficiencies, this project aimed to identify key parameters for developing a predictive system that could better manage surges by utilizing leading indicators and improving patient discharge processes.

Through comprehensive engagement with stakeholders across different departments, including shadowing key personnel, we identified discharge inefficiencies as a primary contributor to surge-related issues. By refining the problem statement and using a problem definition worksheet, we pinpointed root causes and outlined a roadmap for improving both patient flow and surge management. Our recommendations focus on enhancing the predictive model by incorporating leading indicators, such as discharge rates and bed turnover, to forecast surges 12 hours in advance. This proactive approach will allow the hospital to optimize resource allocation, improve patient throughput, and reduce reliance on ambulatory bypass, ultimately improving patient outcomes and hospital operations.

In the short term, our recommendations prioritize real-time monitoring of key parameters, such as patient inflows from the ED, bed availability, and discharge processes. For long-term improvements, the hospital should expand its predictive capabilities by incorporating historical admission data, external factors such as seasonal trends, and enhanced capacity and resource management strategies. Implementing these recommendations along with the framework which we suggested will create a more responsive and efficient healthcare environment, ensuring that Bayview Medical Center can effectively manage patient surges and improve the quality of care provided.

Introduction & Background

Bayview Medical Center is currently facing significant operational challenges due to surges in patient volume, particularly in the Inpatient Medicine Department, which accounts for 50% of the hospital's bed capacity. Despite the implementation of Active Bed Management (ABM) strategies and advancements in electronic health records, the hospital is experiencing an alarming rise in ambulatory bypass instances. This increase in patient inflow has led to critical throughput issues, where the number of patients awaiting care exceeds the available beds, necessitating immediate intervention.

Problem Statement:

This project aims to develop a predictive model for managing patient surges on the Medicine Floor by identifying key parameters such as Incoming patient demand, Hospital capacity, discharge rates, average length of stay (LOS), and external covariates. The model will forecast surges 12 hours in advance to enhance prediction accuracy, optimize resource allocation, and ensure timely patient discharges while minimizing wait times.

Why Is it a Problem?

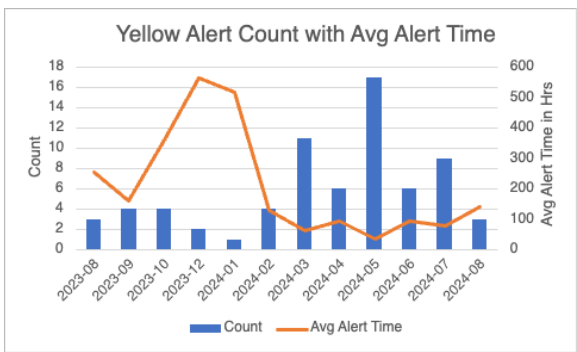


Figure 1^[3]

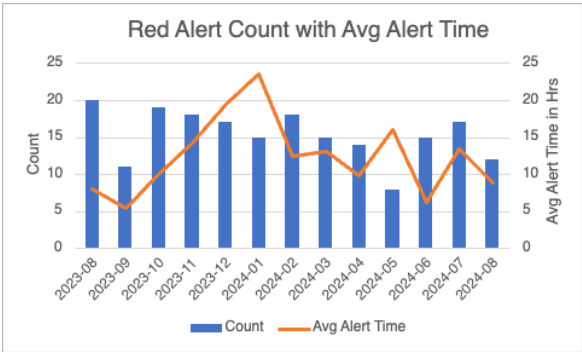


Figure 2^[3]

The impact of these surges is profound. For instance, Figure 1 illustrates the frequency of Yellow Alerts, which indicate when the Emergency Department (ED) is overwhelmed with incoming patients. These scenarios create bottlenecks that hinder the timely transfer of patients to the Medicine Department, resulting in extended wait times and potential delays in care. Furthermore, Figure 2 depicts Red Alerts, signaling that the Intensive Care Unit (ICU) is at capacity, preventing the admission of critically ill patients. Such situations can lead to increased stress for both staff and patients and may affect overall health outcomes.

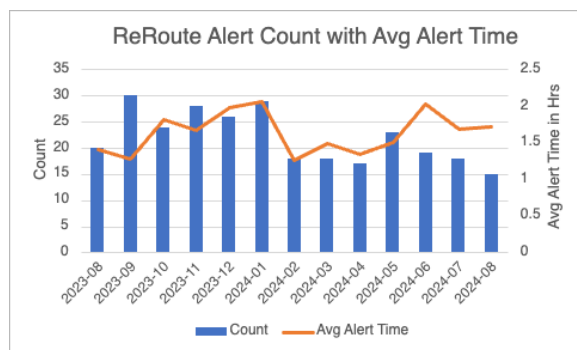


Figure 3^[3]

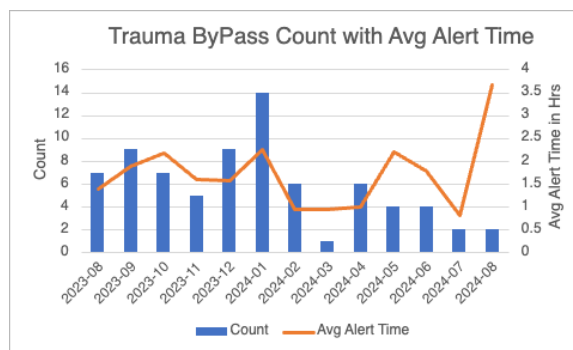


Figure 4^[3]

Additionally, Figure 3 shows Re-route Alerts, which occur when ambulances are directed to alternate facilities due to a lack of available beds. This not only impacts patient care but also represents a missed opportunity for the hospital to serve those in need. Similarly, Figure 4 highlights Trauma Bypass instances, where incoming trauma patients must be transferred to other hospitals because Bayview cannot accommodate them. These challenges threaten the hospital's capacity to provide timely, high-quality care.

In response to these systemic challenges, our project aims to develop a predictive model that leverages leading indicators to manage patient surges more effectively. The current model relies primarily on trailing indicators, leading to delayed responses to surges and not accounting for critical backend factors such as patient discharge rates. By identifying key parameters that can predict surges in patient volume, the Medicine Department can proactively manage bed availability, optimize resource allocation, and enhance discharge planning.

This initiative seeks to address the fluctuating patient volumes in the Medicine Department, driven by unpredictable inflows from the ED and other sources. By focusing on forecasting and strategic planning, the project aims to create a more responsive healthcare environment, ultimately leading to improved outcomes for patients and a more sustainable operational model for Bayview Medical Center.

Methodology

This project follows a structured methodology starting with Stakeholder Engagement and Process Mapping to understand current workflows. Next, Data Analysis and Visualization help identify key trends, followed by a Literature Review and Best Practices Research to incorporate industry standards. Leading Indicators are then identified to guide predictions, leading to Model Development. Finally, the methodology concludes with Implementation Planning to ensure effective integration of the model into hospital operations.

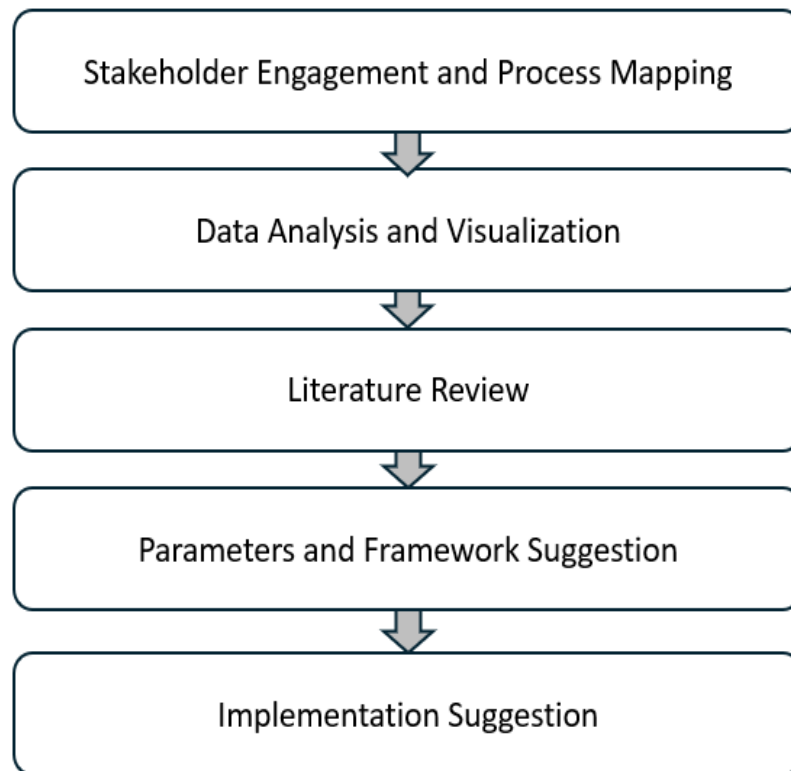


Figure 5

Stakeholder engagement and process organization: Through interviews and follow-up observations with emergency department staff, the project team gained insight into the existing processes, identifying major pain points and improvement needs. Next, during the data analysis and visualization phases, the team used Tableau to analyze key indicators, including length of stay (Los) and discharge rate, and to identify long-term trends by establishing baseline performance indicators.

Data analysis and visualization: Using the Tableau Dashboard, the project team analyzed key indicators within the emergency department, such as length of stay (Los) and discharge rates. Through the visualization of these data, the team established baseline performance indicators, identified historical trends, and provided data support for subsequent forecasts.

Literature Review: The team looked at existing research and models for predicting patient surges, drawing on experience and methodology. The team then moved on to the identification of forward-looking indicators, proposing measurable parameters to achieve more accurate surge forecasting, shifting the focus from passive management to active prevention.

Parameters and Framework Suggestion: In this phase, the team developed both short- and long-term recommendations to guide the hospital in building its predictive model. Additionally, we proposed a comprehensive framework that the hospital can adopt for constructing an effective and robust predictive model.

Implementation Suggestion: It is the key stage after model development. The team made specific, actionable suggestions for improvement based on the short term and long-term parameters. The recommendations focus on how to optimize hospital bed management and improve patient mobility, with the goal of improving overall operational efficiency and ensuring an effective response to patient surges.

Analysis

Original Problem Statement:

This project's main objective is to identify the parameters of a predictive system based on leading indicators to optimize patient surge management in the Emergency Department (ED) at inpatient medicine services. The current model, which relies on trailing indicators and lacks consideration of patient discharge rates can be improved to increase the prediction accuracy.

Stakeholder Interview:

After going over the materials provided by the host and other sources, we held interviews with key stakeholders to get a better understanding of the patient flow in the hospital and find parameters which can be used in the model. We also wanted to see if the real-life challenges matched what was described in the literature. By talking to people involved in different parts of the process, we were able to gather insights from their unique perspectives, which helped us identify the parameters.

Insights:

Through our discussions and observations with key stakeholders across the Medicine Department, we gained a comprehensive understanding of the patient flow processes and the complexities involved in managing admissions, bed assignments, and discharges, all of which contribute to the surge management challenges faced by the department.

- **Medicine Triage:**

Shadowing Dr. Kiyatkin provided critical insights into the medicine triage process, where the admission of patients is based on an assessment of the patient's health status and criticality. We observed how medical urgency, available resources, and specific departmental protocols dictate the decision to admit patients to the hospital floor. This gave us a clearer picture of the points of contact in the triage process and the key considerations that must be weighed in determining patient flow from the ED to the Medicine Department. This understanding laid the groundwork for identifying potential bottlenecks in the initial stages of patient admission.[5]

- **PCC Bed Assignment:**

Our observation with Kathleen O'Brien shed light on how the Patient Care Coordinators (PCC) manage bed assignments and optimize capacity within the hospital. We learned about the different operational protocols followed by the PCC, specifically distinguishing between the responsibilities of House staff and those using the CIMS system. Kathleen walked us through the decision-making processes involved in allocating beds, emphasizing how different departments communicate to ensure optimal bed utilization, particularly under high patient volume conditions, such as during the COVID pandemic. This provided us with an understanding of how bed allocation is a key determinant in managing patient flow and preventing bottlenecks.[10]

- **Provider:**

Dr. Amy Yu provided in-depth knowledge about how discharge orders are processed through the EPIC system and how this impacts patient flow. Discharge processes depend on the patient's clinical readiness and external factors such as transportation (ambulance availability), family readiness, and coordination with post-hospital care. This multilayered process often results in delays in discharges, which in turn affects bed availability for incoming patients. Dr. Yu also emphasized the categorization of patients based on their condition, which helps prioritize discharges and manage resources effectively. Understanding these dynamics allowed us to recognize the critical role that discharge efficiency plays in alleviating surges.[6]

- **Charge Nurse:**

Leilani Turman's input provided us with a detailed perspective on how Charge Nurses manage the discharge process and the various internal and external factors that contribute to delays. We observed how internal hospital coordination, such as communication between nursing staff, providers, and case management, interacts with external factors like family coordination and post-discharge transportation. Leilani's explanation helped us identify the significant role Charge Nurses play in managing patient flow on the ground and

how improving these processes could alleviate surges on the Medicine Floor by speeding up patient discharges.[12]

- **Patient Care Manager:**

Ms. Yiankos offered insights into the discharge process from the perspective of nursing staff and the Patient Care Management team. She identified several areas where internal communication could be improved, such as streamlining the flow of information between different departments and addressing the delays caused by miscommunication or unclear discharge instructions. This understanding was reinforced by Ms. Yiankos's emphasis on the need for better coordination across various roles to prevent discharge delays and, subsequently, manage bed availability more effectively.[13]

- **Nursing Department, Case Management, and Social Work:**

Through our participation in an interdisciplinary roundtable meeting involving providers, charge nurses, case managers, and social workers, we observed how critical information is exchanged across these roles. We witnessed how they work together to predict upcoming discharges and plan for patient transitions. This meeting highlighted the collaborative nature of discharge planning, where each role contributes to forecasting patient movement, making it clear that early identification of potential discharges is essential for managing surges. We also gained insight into the influence of social workers in coordinating post-discharge care, further emphasizing the importance of cross-departmental collaboration.

- **Dr. Gundareddy (Host, Medicine Department):**

Throughout our project, we had multiple meetings with Dr. Gundareddy, each helping us refine our approach and deepen our understanding of the operational challenges faced by the Medicine Department. In our initial discussions, Dr. Gundareddy played a crucial role in helping us define and clarify the problem statement, ensuring that our focus remained on addressing the surges in patient volume specific to the Medicine Department. He also provided a detailed overview of the Emergency Department (ED) process, highlighting how patient inflows from the ED contribute significantly to surges in the Medicine Department and creating downstream effects on bed availability and patient throughput. As we progressed with our project, Dr. Gundareddy offered continual feedback on our work, giving us insights on the evolving nature of the surge problem and helping us adjust our focus where necessary. His input was particularly valuable in identifying critical parameters that could serve as leading indicators for predicting surges, allowing us to consider not only patient inflows but also the role of discharge processes, bed turnover rates, and the coordination of resources. Dr. Gundareddy's guidance was instrumental in ensuring that we maintained a strong clinical and

operational perspective as we worked through our analysis, enabling us to consider both the mathematical and practical aspects of the problem.[4]

Redefined Problem statement:

Through extensive engagement with key stakeholders across the patient flow process from triage and admission to discharge and bed management. We developed a comprehensive understanding of the factors influencing the Medicine Department's ability to effectively manage patient surges. These varied perspectives provided valuable insights into both internal and external variables impacting surge management, helping us identify discharge as the primary factors contributing to the issue. By utilizing a problem definition worksheet, impact matrix (Appendix 2) and issue tree (Appendix 1), we refined the scope, pinpointed root causes, and adjusted our problem statement to more accurately address these underlying issues. This will guide the development of our predictive model and recommendations for process improvements.

This project aims to develop a predictive model for managing patient surges on the Medicine Floor by identifying key parameters such as Incoming patient demand, Hospital capacity, discharge rates, average length of stay (LOS), and external covariates. The model will forecast surges 12 hours in advance to enhance prediction accuracy, optimize resource allocation, and ensure timely patient discharges while minimizing wait times.

Recommendations

After Analyzing the process and considering all the factors our team has come with 3 recommendations for the hospital:

Short Term Recommendation: Predicting Surge 12 Hrs in Advance

For predicting surges 12 hours in advance, we recommend focusing on the following five key parameters. These parameters are crucial for real-time monitoring and quick decision-making to manage patient flow effectively.

1. Patient Incoming from ED to Medicine

Explanation: This parameter tracks the number of patients being transferred from the Emergency Department (ED) to the Medicine department within the hospital. It indicates the immediate demand for beds and care in the Medicine unit.

Impact: Monitoring this parameter closely helps identify surges in real time, as an increase in transfers from the ED typically signals a spike in patient load that the Medicine floor must accommodate.

Data Source: Hospital Information System (HIS), Dashboard

2. Net Internal Patient Transfers

Explanation: This refers to the movement of patients within the hospital, such as transfers from one department to another. Net internal transfers can redistribute the patient load and help balance capacity.

Impact: Tracking internal transfers ensures that the hospital efficiently utilizes available beds across departments, reducing bottlenecks and preventing overcapacity in any single department.

Data Source: Dashboard, Bed Management System.

3. Beds Available

Explanation: Real-time data on the number of available beds in the Medicine department and other units is crucial for managing patient flow during a surge. This parameter provides a snapshot of how much capacity remains.

Impact: Knowing the number of available beds allows hospital staff to make informed decisions about whether to admit new patients or transfer existing patients to other departments with more space.

Data Source: Bed Management System.

4. Number of Discharges

Explanation: This parameter tracks the number of patients being discharged from the Medicine department. It provides insight into how quickly beds are becoming available for new patients.

Impact: Monitoring discharges helps forecast bed availability and manage patient flow effectively, particularly during high-demand periods. Faster discharge processes improve bed turnover and reduce strain on capacity.

Data Source: Dashboard

5. Mortality

Explanation: The number of deaths occurring in the hospital or Medicine department, while a sensitive metric, can impact capacity planning. Mortality events result in bed availability, and tracking this parameter helps ensure timely responses.

Impact: Monitoring mortality helps predict sudden capacity availability, which can be factored into overall bed management during a surge.

Data Source: Dashboard

$$\text{Surge Likelihood} = \frac{\text{Patient Incoming from ED} + \text{Patient Transfer In}}{\text{Beds Available} + \text{No. of Discharges} + \text{Mortality} + \text{Patient Transfer Out}}$$

What Does the Values of Surge Likelihood Mean -

Surge Likelihood > 1 - Surge is Imminent

Surge Likelihood > 0.8 - Surge May Happen

Surge Likelihood < 0.8 - No Surge

To account for surge likelihood in the Medicine Department, a 20% leeway is being considered, based on the observation that approximately 20% of patients admitted to the Medicine Department come from the Emergency Department (ED). This leeway allows for more accurate surge predictions, reflecting the typical patient inflow from the ED.

How To Use Surge Likelihood Formula-

To effectively manage patient flow, the Surge Likelihood should be calculated daily, with specific evaluations conducted at 5 AM and 5 PM. This dual-timed calculation can be leveraged independently by each department, as well as in a combined format to provide a comprehensive overview of departmental and overall capacity.

The results from these calculations enable proactive analysis, allowing us to identify actionable strategies to optimize patient flow. For instance, we can revisit discharges scheduled for the following day to determine if additional patients can be safely discharged. Moreover, if the Medicine Department is nearing capacity while the Emergency Department (ED) is not in a surge situation, we can coordinate with the ED to expedite discharges from there, alleviating pressure on the Medicine Department.

By analyzing input data for Surge Likelihood over multiple days, we can uncover throughput challenges. This insight will allow us to target specific areas for improvement, ensuring we enhance operational efficiency and better serve our patients.

Long Term Recommendation: Predicting Surge Beyond 12 Hrs

The hospital can use long-term recommendations to predict surges beyond 12 hours in advance. The demand, capacity & resource, and Discharge parameters can be used in the framework given below to decrease the discharge timings, reduce patient time, and improve resource allocation. These enhancements will enable better decision-making, ensuring that the hospital can proactively manage patient flow and maintain optimal operational efficiency.

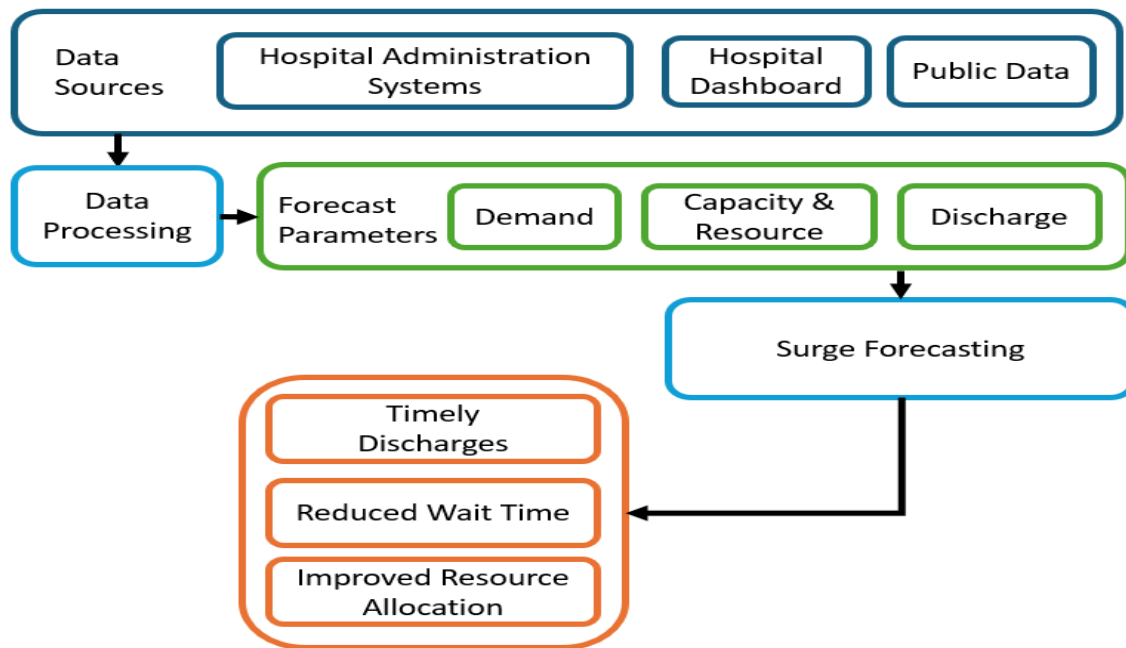


Figure 6^[11]

Framework Overview:

The primary component of the framework and the focus of this work is to forecast surges (Figure 6, Blue). It is data-driven and prospective, so it requires data from the hospital system (Figure 6, Green) and a variety of data from sources (Figure 6, Dark Blue). This will eventually help in optimizing (Figure 6, Orange).

Based on literature review, we recommend hospitals use the Time Series Dense Encoder (TiDE) for demand forecasting. Here's why TiDE is particularly suited for this task:

- **Encoder-Decoder Architecture:** TiDE's encoder-decoder architecture is highly effective in identifying complex patterns in historical hospital data and translating them into accurate predictions of future surges. This allows hospitals to anticipate demand spikes with greater precision. Encoder looks at the past data and finds some historical trends and converts them into a

mathematical representation. On the other hand, the decoder that mathematical representation and uses it to do prediction.[11]

- **Flexible Forecasting:** It can handle various input covariates (such as time of day, past admission rates, seasonal patterns, or external factors like weather) and can generate predictions for different horizons. Moreover, TiDE can be easily adapted to different hospitals or regions with minimal adjustments.[11]
- **Uncertainty Quantification:** TiDE supports uncertainty estimation by performing quantile regression. This means it doesn't just predict a single value, but a range of possible outcomes (with confidence intervals), which is crucial for robust decision-making in uncertain environments like hospital surge management. [11]

Forecast Parameters

Below is a list of forecast parameters that hospitals can incorporate into the surge forecasting framework, along with their potential impacts and sources for data collection:

1. Demand Related Parameters:

a. Historical Patient Admissions

Explanation: Past data on patient admissions can reveal patterns and trends that allow the hospital to anticipate future patient inflow. For example, historical data can reflect the demand during specific times of the year (e.g., flu season), days of the week, or external events that impact admissions.

Impact: By analyzing historical admission patterns, the hospital can improve demand forecasting accuracy, which helps in managing resources and preparing for surges.

Source: Dashboard/Hospital Information System (HIS).

b. Transfer from other facilities/clinics

Explanation: Tracking the number of patients transferred from external healthcare facilities, including nearby hospitals, clinics, or long-term care centers, can provide insights into the expected patient inflow during specific periods.

Impact: Understanding transfer patterns allows hospitals to predict and plan for additional patient volumes, particularly during surge periods, and ensure adequate staffing and bed availability.

Data Source: Dashboard.

c. Seasonal Data/External factors

Explanation: Seasonal variations, weather conditions, and public health events such as pandemics or flu outbreaks can influence hospital admissions. External data such as temperature, public event schedules, or epidemiological trends can be incorporated into forecasting models.

Impact: Including these factors in the forecasting model allows the hospital to anticipate surges driven by external conditions, leading to more proactive resource allocation.

Data Source: Public data (e.g., weather services, health agencies, Google Trends).

2. Capacity and Resource Management Parameters:

a. Hospital Capacity

Explanation: The total number of available beds across different departments (e.g., ICU, general wards, emergency) defines the hospital's ability to accommodate patients. Monitoring and updating this data regularly are critical to managing capacity effectively.

Impact: Knowing the exact capacity helps the hospital avoid overbooking or underutilization of beds, ensuring that resources are matched to the predicted patient load.

Data Source: Dashboard.

b. Current Bed Occupancy

Explanation: Real-time data on bed occupancy shows how many beds are currently in use and which departments are nearing or exceeding their capacity.

Impact: Monitoring occupancy rates helps identify departments that might face strain during a surge, allowing for the reallocation of resources or the activation of additional beds.

Data Source: Dashboard

c. Resource Availability

Explanation: This includes tracking the availability of critical resources such as staff (nurses, technicians, doctors), medical equipment (e.g., ventilators, monitors), medications, and other supplies. Having real-time data on resource availability enables better planning and response during patient surges.

Impact: Ensuring resource availability improves the hospital's ability to handle more patients, especially during surges, and prevents bottlenecks in patient care due to resource shortages.

Data Source: Dashboard, Inventory Management System.

3. Discharge Parameters:

a. No. of Patients to be Discharged

Explanation: The number of patients ready for discharge indicates which beds will soon become available. Tracking discharge status in real time allows for more efficient bed management.

Impact: Predicting discharges accurately ensures that beds are turned over quickly and are available for incoming patients, particularly during high-demand periods.

Data Source: Dashboard.

b. Length Of Stay

Explanation: The expected length of stay for patients can be modeled using statistical methods like Maximum Likelihood Estimation (MLE) based on historical data. This helps predict when patients are likely to be discharged and when beds will be freed up.

Impact: Predicting LOS accurately improves bed management and resource planning by ensuring that capacity aligns with patient needs and minimizing unnecessary bed blockages.

Data Source: Dashboard

c. Bed Turnover

Explanation: Bed turnover refers to the time taken to clean and prepare a bed after a patient is discharged before the next patient can be admitted.

Impact: Reducing bed turnover time maximizes the hospital's efficiency in admitting new patients during peak demand periods, thereby increasing capacity utilization.

Data Source: Dashboard or Bed Management System.

d. Discharge Process Completion - Pending Diagnostic or Treatment Results

Explanation: Discharge delays often occur due to pending diagnostic results or treatments that need to be completed before a patient can leave. Monitoring the completion of diagnostics and treatments helps expedite the discharge process.

Impact: Timely completion of diagnostic tests and treatments facilitates faster patient discharges, which in turn helps free up beds for new patients, particularly during surges.

Data Source: Dashboard, Laboratory Information System (LIS), Radiology Information System (RIS).

Future Implementation: Expanding the Framework for Capacity Management Optimization

This framework can be further developed into a capacity management optimization model, providing Bay View Hospital with recommendations for optimal capacity allocation and patient transfers.

Other Recommendation

1. Discharge Lounge

Explanation: A discharge lounge is a dedicated area where patients who are medically ready to leave the hospital but are awaiting transportation or final paperwork can stay temporarily.

Impact: This reduces bed blockage by allowing patients to leave their hospital bed and move to the discharge lounge, freeing up the bed for the next patient while ensuring the patient can be discharged efficiently.

2. Set notification for Discharge orders

Explanation: Nurses are often occupied with patient care and may not always have immediate access to the computer system to check whether a discharge order has been placed. Implementing a notification system within the EPIC software that sends push notifications when a discharge order is issued would be highly beneficial. This would allow nurses to stay updated without needing to be at their workstation.

Impact: This system would improve nurses' awareness of upcoming bed availability, enabling them to manage resources more effectively and streamline patient discharge processes.

3. Input details post inter-departmental (ID) meetings into EPIC

Explanation: During the ID meetings, patient discharges are discussed and planned based on their condition, but these details are often not entered into the EPIC system. This creates a communication gap between doctors and other team members, as they are not updated with the meeting outcomes.

Impact: Assigning someone the responsibility of entering discharge planning details from the ID meetings into EPIC would bridge this communication gap.

It would lead to better coordination among doctors and staff, improving overall discharge planning and execution

4. Prioritization of Patient Visits for Technicians and Consultants

Explanation: The primary focus for technicians and consultants is to attend to patients in the ED. Once those patients are seen, it can be challenging to determine which department to prioritize next, often leading to delays in discharges as patient visits in other departments are not effectively scheduled.

Impact: Implementing a priority list for technicians and consultants, based on patients who are likely to be discharged soon, would improve discharge management. This approach could help reduce delays in discharges by ensuring that patients who are nearing discharge receive timely attention.

Conclusion

Optimizing patient surge management is essential for improving patient outcomes and ensuring efficient use of hospital resources. By shifting from reactive, trailing indicators to a predictive model based on leading indicators, Bayview Medical Center can better anticipate patient surges, reduce strain on the Medicine Department, and minimize instances of ambulance bypass. Incorporating real-time data on patient inflows, discharge processes, and bed availability will enable faster and more informed decision-making, allowing for better resource allocation and improved patient throughput.

The recommendations provided in this report offer both immediate and long-term solutions to enhance surge management. In the short term, real-time monitoring of key parameters will provide actionable insights, while long-term strategies, such as analyzing historical admission trends and external factors, will further strengthen the hospital's predictive capabilities. By adopting these changes, Bayview can improve its operational efficiency, reduce patient wait times, and ultimately provide higher-quality care, positioning itself as a more resilient and responsive healthcare provider.

Implementing these strategies will create a sustainable framework for managing capacity, enabling Bayview Medical Center to not only address current challenges but also be better prepared for future surges in patient volume.

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Appendix

1. Factors Affecting Discharge Process (Issue Tree)

The issue tree was created to comprehensively identify all factors impacting the discharge process. This approach helped us to systematically narrow down the problems, allowing us to focus on the key issues that were both within our scope and critical to the hospital.

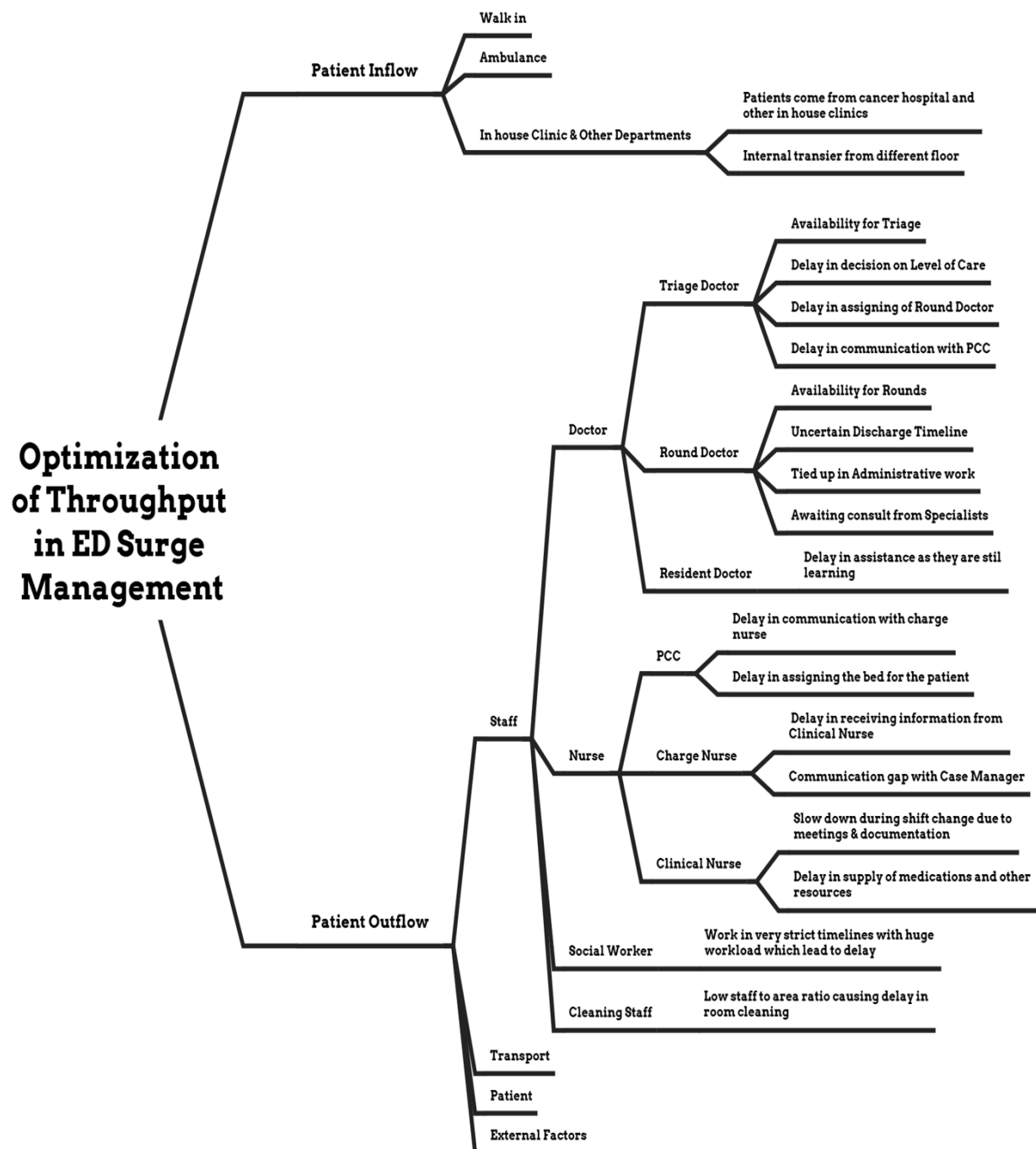


Figure 7

2. Impact Matrix for recommendations

The impact matrix was used to categorize the key problems identified from the issue tree into a 2x2 matrix, based on their impact and effort required. This approach allowed us to prioritize and address the high-impact issues that could be resolved within the given timeframe.

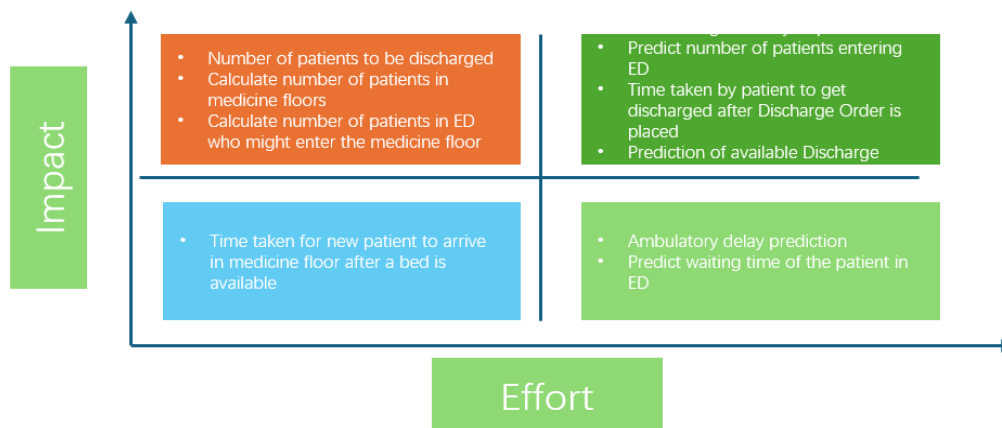


Figure 8

3. Patient Flow

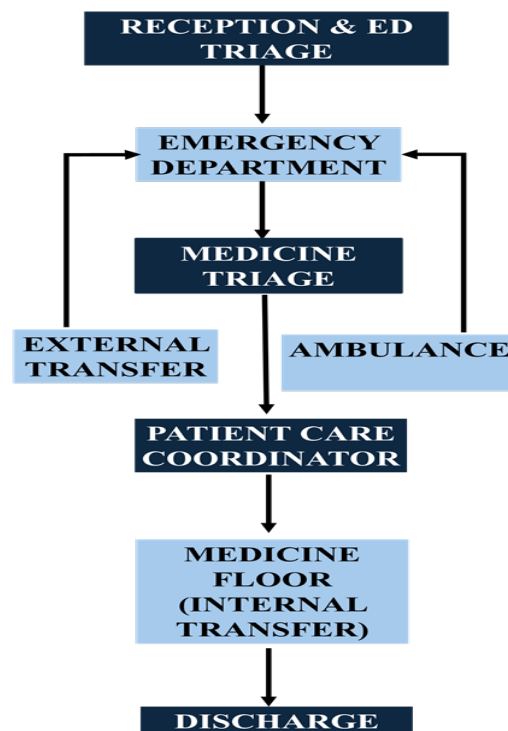


Figure 9

The Medicine Department frequently faces significant fluctuations in patient volume due to unpredictable inflows from various sources, including the Emergency Department (ED) and internal transfers from other units. These sudden surges in-patient admissions exert substantial pressure on the department's capacity to manage bed availability, leading to longer wait times, overcrowding, and extended lengths of stay (LOS). In detail the patient flow in the hospital is:

1. **Reception & ED Triage:** When a patient enters the hospital, they are first assessed by the triage team to determine whether emergency department (ED) care is required. If so, the team evaluates the level of urgency to prioritize treatment accordingly.
2. **Emergency Department:** In the ED, the patient is initially stabilized. Patients arrive through various channels, including walk-ins, ambulance services, and external transfers from other hospitals.
3. **Medicine Triage:** The medicine triage team assesses whether the hospital's medicine department can provide the appropriate level of care for the patient. They also assign a provider to manage the patient's care.
4. **Patient Care Coordinator (PCC):** The PCC evaluates bed availability and assigns a bed based on factors such as the patient's gender and specific medical condition. PCC also assigns the charge nurse to the patient.
5. **Medicine floor (Internal Transfer):** Due to unforeseen circumstances, a patient may need to be transferred internally within the medicine floor to receive the necessary care.
6. **Discharge:** Once the patient is ready for discharge, an order is placed. The actual discharge may take anywhere from a few hours to a few days, depending on whether the patient is returning home or being transferred to another facility.